

Experiment v lingvistice

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ARTS

- a talk on meaning and its acquisition in Czech
- formal semantics provides tools for modeling the meanings of linguistic expressions and their relations
- experimental linguistics lets us test how adults interpret language
- acquisition experiments show how semantic representations develop in children (initially non-adult-like, but gradually converging on adult grammar)

*4.002 Man possesses the capacity of constructing languages, in which every sense can be expressed, without having an idea how and what each word means.
Tractatus Logico-Philosophicus, Ludwig Wittgenstein (Ogden translation, 1922)*

Outline:

1. problem and roadmap: distributivity and the expressions that encode it
2. data and analysis: an experimental study of how universal quantifiers and bare NPs are acquired in Czech
3. conclusion: a theoretical explanation of the results and predictions for future research
4. placing the results in the broader context of my research

Problem

- children often make errors during language acquisition
- but according to a standard textbook generalization (see Pinker 1995 etc.), phonological, morphological, and syntactic acquisition is largely completed before children enter elementary school
- however, semantic (and pragmatic) acquisition takes much longer
- one key issue is the acquisition of distributivity:

(1) Dva autoři napsali tři knihy.

- a. distributivní čtení: _A → , , ; _B → , , 
- b. kumulativní čtení: _A → , ; _B → 
- c. kolektivní čtení: _A + _B → , , 

Some expressions (singular universal quantifiers, e.g. *each*, *jedes*, *každý*, ...) are obligatorily interpreted as distributive in adult grammar:

(2) Každý autor napsal tři knihy.

- a. distributivní čtení:  $A \rightarrow$ , , ;  $B \rightarrow$ , , 
- b. #kumulativní čtení:  $A \rightarrow$ , ;  $B \rightarrow$ 
- c. #kolektivní čtení:  $A +$  $B \rightarrow$ , , 

- see the slide with the adult control group (N=47) in the appendix
- however, not in children's grammar

- similarly, while singular universal quantifiers necessarily prefer a distributive reading, Czech bare plural noun phrases strongly prefer a non-distributive reading (= cumulative or collective):

(3) Každé dítě si koupilo dva bonbóny.

a. distributivní čtení: _A → , ; _B → , 

b. #kumulativní čtení: _A → ; _B → 

(4) Děti si koupily dva bonbóny.

a. ??distributivní čtení: _A → , ; _B → , 

b. kumulativní čtení: _A → ; _B → 

- an experiment focused on cumulative vs. distributive readings of both types of expressions

- there is now a substantial literature showing that the target interpretation of SUQs is acquired relatively late: Geurts (2003); Musolino (2009); Pagliarini et al. (n.d.); Koster et al. (2018) etc.
- studies on Italian, English, and Dutch
- distributivity is one of the core problems
- based on this literature, the first research question is:

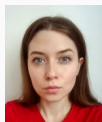
(5) Research question 1: How does acquisition of the distributive interpretation of SUQs develop in Czech?

- given the specifics of Czech (previous studies contrasted SUQs with collective readings of definite NPs), the second research question is:

(6) Research question 2: How does acquisition of cumulative vs. distributive readings of bare NPs develop in Czech?

- Slavic languages do not have a grammaticalized definiteness/indefiniteness contrast (see Šimík and Demian 2020), but they are specific in allowing bare NPs in both plural and singular
- most previous studies focused on the acquisition of singular SUQs contrasted with definite NPs ... languages with a developed definiteness/indefiniteness system (Italian, English, Dutch) vs. languages without this system (Czech, most Slavic languages)

- SUQs are part of the quantificational system of a language, and their linguistic properties (including acquisition) may differ depending on how the rest of that system is structured
- a general (repeatedly validated) assumption: acquisition of semantics precedes acquisition of pragmatics (Syrett 2018; Noveck 2001; Papafragou and Musolino 2003; Katsos and Bishop 2011 etc.)
 - e.g. the exhaustified interpretation of existential quantifiers (Noveck 2001) is still non-target-like even around age 10 (pragmatics)
 - similarly for the exclusive interpretation of logical disjunction
 - more complex semantic relations are acquired between ages 7 and 11 (temporal relations between sentences, ...: Syrett 2018)
- work on the experiment together with Žaneta Šulíková & Martin Juřen:
data and description: Šulíková (2023), preliminary analysis: Dočekal and Šulíková (2024)



- 214 Czech children (age range: 5–11 years, median age: 7) completed the filler items in the experiment
 - on a tablet (in preschools and elementary schools)
 - pilot study: around 30 children, design changes
 - shortened (without context), phrased more directly
 - pilot data were not used
- 47 Czech adults (median age: 24) also completed the filler items (online in PCIBex)
 - shared via link, on PC (demo link, source code)

Design

- 2×2 design:
 - a universal quantifier (UNIV) or a bare NP (BARE) was tested against
 - a distributive (DISTR) or cumulative (CUMUL) interpretation (pictures) in a standard Latin-square design

- (7) a. Každé dítě si koupilo dva bonbóny. UNIV
b. Děti si koupily dva bonbóny. BARE

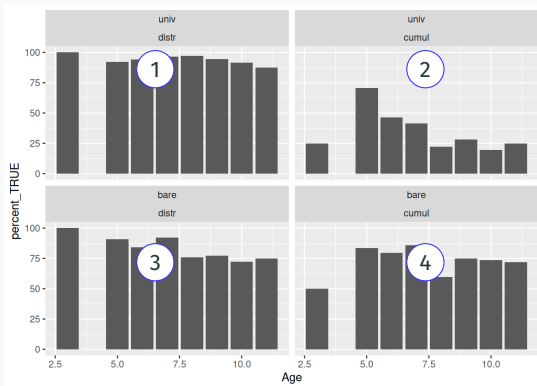


Obrázek 1: Distributive interpretation



Obrázek 2: Cumulative interpretation

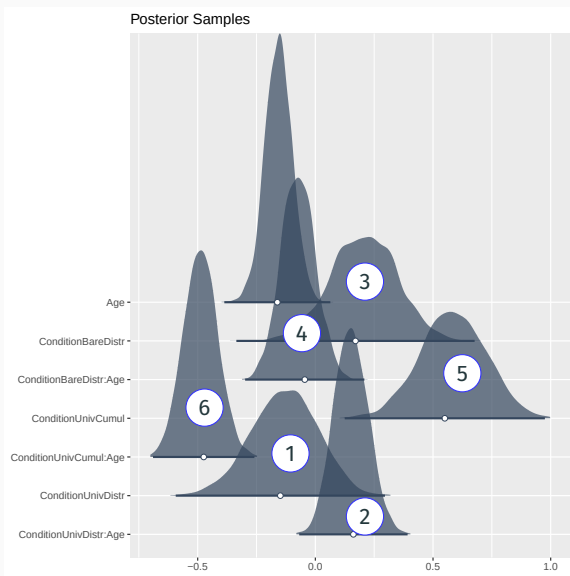
Descriptive Statistics



Obrázek 3: Development of acquisition in children (N=214)



Obrázek 4: Adult control group (N=47)



Obrázek 5: Bayesian model for children

The Bayesian model for children shows that:

1. the cumulative interpretation of bare NPs, (1) (reference level), is as acceptable as the distributive interpretation of universal quantifiers
 - this does not change, (2) during acquisition: non-credible interaction UNIVDISTR x AGE
 - in line with adult grammar
2. the distributive interpretation of bare NPs, (3), is accepted across the whole child sample (non-credible main effect BAREDISTR – relative to the reference level) – unlike adult grammar
 - during acquisition this does not change, (4): non-credible interaction between BAREDISTR and AGE
 - mismatch with adult grammar

3. the cumulative interpretation of universal quantifiers is accepted across the whole child sample, (5) (credible main effect UNIVCUMUL – relative to the reference level)
 - however, this changes during acquisition, (6): credible negative interaction between UNIVCUMUL and AGE – development toward adult grammar

- Czech children (aged 5–11) gradually acquire the correct interpretation of universal quantifiers (UQs)
- at first, they incorrectly accept both distributive and cumulative interpretations
- eventually, however, they reach an adult-like grammar in which they reject the latter interpretation (answer to question 1):

(8) Research question 1: How does acquisition of the distributive interpretation of SUQs develop in Czech?

- a. výchozí fáze: chybná interpretace i nedistributivního čtení UQ (5.–8. rok – ve vzorku)
- b. finální fáze: správná interpretace UQ (9.–11. rok)

- however, for bare plural NPs the situation is different: even at age 11, children accept the distributive interpretation to the same extent as the cumulative one

(9) Jak probíhá vývoj osvojování kumulativního vs. distributivního čtení holých NP v češtině?

a. ve zkoumaném vzorku (5–11 let) děti přijímají i (vůči gramatice dospělých) chybnou, tj. distributivní interpretaci holých NP v plurálu

- likely answer: this is related to children acquiring semantics first and only later the pragmatics of individual expressions

Theoretical Explanation

- a pragmatic-semantic explanation; application of standard steps (Horn 2001; Grice 1975 etc.) in adult grammar:
 - (10): an adult understands that (10-a) is necessarily distributive
 - (10): it is an ambiguous sentence (it could also have a distributive reading), however (10-a) is logically stronger than (10)
 - an adult assumes that a cooperative speaker would use (10-a) for a distributive interpretation, not ambiguous (10)
 - → therefore, an adult treats (10-a) as a false interpretation of (10)
- pragmatics as comparison of what is said with what could have been said (alternatives) – it is generally known that children acquire this type of pragmatic reasoning relatively late (see Sauerland and Yatsushiro 2018; Syrett 2018 etc.)

(10) Děti si koupily dva bonbóny.

a. Každé dítě si koupilo dva bonbóny.

Preliminary Summary

1. Starting problem: one mismatch between child and adult grammar (distributivity)
2. Data: Czech provides (besides standard SUQs) bare NPs: both expression types show a different acquisition profile (experiment)
3. Conclusion: partial confirmation of previous research on acquisition of SUQ distributivity, plus interpretation of a new finding (later acquisition of the correct non-distributive meaning of bare NPs)

Conclusion and Placement in the Research Context

- projects, outputs, students
- 1. GAČR project „Negation and negative polarity expressions in Czech“, 2007–2009, postdoctoral project
 - 14 publications, with the habilitation thesis “Czech Negation from the Formal Perspective” as another output
 - followed by Hana Strachoňová’s successfully defended dissertation “Czech free-choice indefinites” (2017)
- 2. GAČR project „Formal approaches to verbal prefixation and aspect in contemporary Czech“, 2009–2011, co-investigator, PI Markéta Ziková
 - 27 publications, including “Atoms, groups and kinds in Czech”
- both primarily in theoretical linguistics (formal semantics)

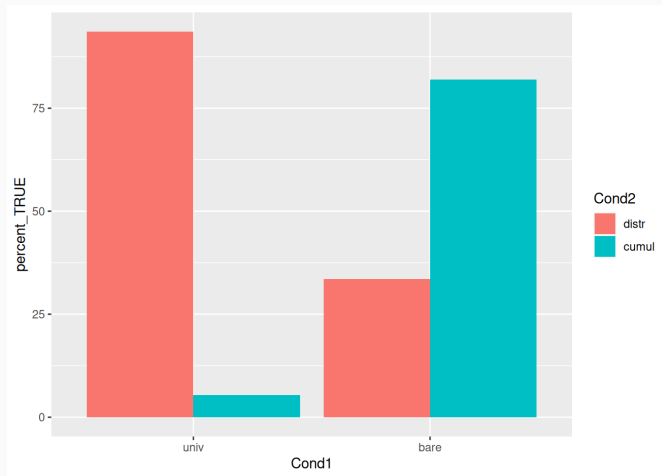
3. GAČR project „Formal approaches to grammatical number in Slavic languages“, 2017–2019, principal investigator

- 58 publications, including “Counting Aggregates, Groups and Kinds: Countability from the Perspective of a Morphologically Complex Language” and “Formal Approaches to Number in Slavic and Beyond”
- this project also enabled the successful dissertation “Subatomic Quantification” by Marcin Wągiel (2019): Dean’s Award, Faculty of Arts MU, E.W. Beth Dissertation Prize 2020

4. MŠMT project „A Whole Life with Language“, PI Pavel Caha, 2025–2028

- thanks to this project: a postdoctoral position for Radim Lacina, a doctoral position for Rhiana Horovská
- (3) and (4) mark a gradual shift toward experimental linguistics, psycholinguistics, and acquisition
- this is reflected (I supervise the MA program) in teaching, in thesis topics developed under my supervision, and in new grant proposals, ...

Thank you for your attention!



Obrázek 6: Adult control group (N=47)

- adult meaning representations of both expression types (UQ and bare NP) are in an entailment relation:

- (11) a. Každé dítě si koupilo dva bonbóny.
 $\forall x \left(\text{Child}(x) \rightarrow \#\{y \in \text{Sweet} \mid \text{Bought}(x, y)\} \geq 2 \right)$
- b. Děti si koupily dva bonbóny.
 $\exists x \exists z (\text{Child}(x) \wedge \text{Child}(z) \wedge x \neq z) \wedge \#\{y \in \text{Sweet} \mid \exists x (\text{Child}(x) \wedge \text{Bought}(x, y))\} \geq 2$
- c. $a \models b$ ale ne naopak: $b \not\models a$

Model

- Bayesian logistic hierarchical model with random effects and default priors was fit using the R package RSTANARM Goodrich et al. (2022):
- the dependent variable was the subject's answer:
 - checkmark: the image corresponds to the sentence vs.
 - cross: the image and the sentence don't correspond
- the independent variables were:
 - the condition (UNIVDISTR, UNIVCUMUL, BAREDISTR, BARECUMUL)
 - age of the subject (in case of children)
 - and their interaction
- the reference level was BARECUMUL

Inferential statistics

A) adults:

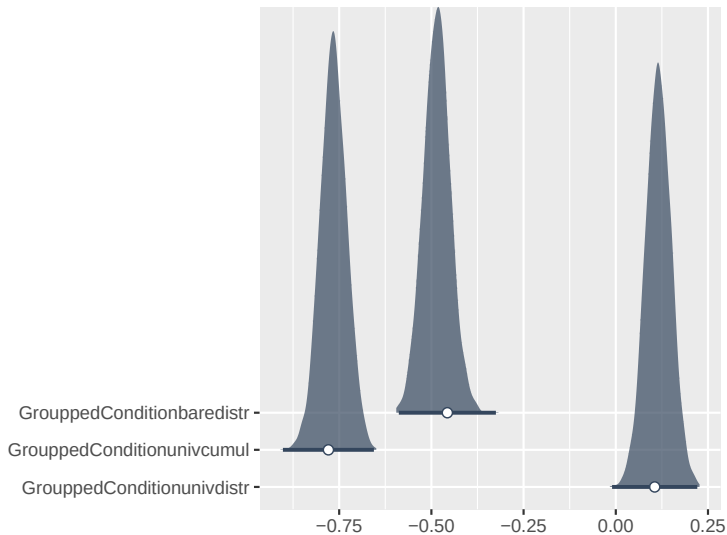
- the distributive interpretation of UQs is acceptable to the same extent as the cumulative interpretation of bareNPs – reference level:
 - $\hat{\mu} = 0.12$, 95% Credibility Interval, CI = $[0.05, 0.18]$, 32.29% in ROPE
- distributive interpretation of bareNPs is much worse than the reference level
 - $\hat{\mu} = -0.48$, CI = $[-0.56, -0.41]$, 0% in ROPE

Inferential statistics

A) adults:

- cumulative interpretation of UQs is also much worse than ref. level
 - $\hat{\mu} = -0.77$, CI = $[-0.84, -0.69]$, 0% in ROPE
- posterior samples graph for adults Bayes model on the next slide

Posterior Samples



Obrázek 7: Bayesian model

Summary of Posterior Distribution

Parameter	Median	95% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	0.82	[0.76, 0.87]	100%	[-0.10, 0.10]	0%	0.999	3640.00
Conditionbaredistr	-0.48	[-0.56, -0.41]	100%	[-0.10, 0.10]	0%	1.000	4237.00
Conditionunivcumul	-0.77	[-0.84, -0.69]	100%	[-0.10, 0.10]	0%	1.000	4170.00
Conditionunivdistr	0.12	[0.05, 0.18]	99.90%	[-0.10, 0.10]	32.29%	1.000	4449.00

Inferential statistics

B) children:

- the distributive interpretation of UQs is the same as the reference level
 - $\hat{\mu} = -0.13$, $CI = [-0.42, 0.13]$, 40.29% in ROPE
- and this remains stable during the acquisition
 - non-credible interaction effect UNIVDISTR by AGE: $\hat{\mu} = 0.15$, $CI = [0.01, 0.30]$, 22.82% in ROPE

Inferential statistics

B) children

- unlike adults, children accept the distributive interpretation of bareNPs to the same extent as the reference level
 - $\hat{\mu} = 0.21$, CI = $[-0.07, 0.47]$, 21.21% in ROPE
- and at least in the sample the development towards adults state is non-credible
 - the interaction between BAREDISTR and AGE: $\hat{\mu} = -0.07$, CI = $[-0.21, 0.07]$, 64.03% in ROPE

Inferential statistics

B) children

- cumulative interpretation of UQs is as a main effect indistinguishable from the reference level
 - $\hat{\mu} = 0.58$, $CI = [0.31, 0.85]$, 0% in ROPE
- but there is a clear development towards adult grammar
 - the interaction between UNIVCUMUL and AGE is a strong negative effect:
 $\hat{\mu} = -0.49$, $CI = [-0.63, -0.34]$, 0% in ROPE

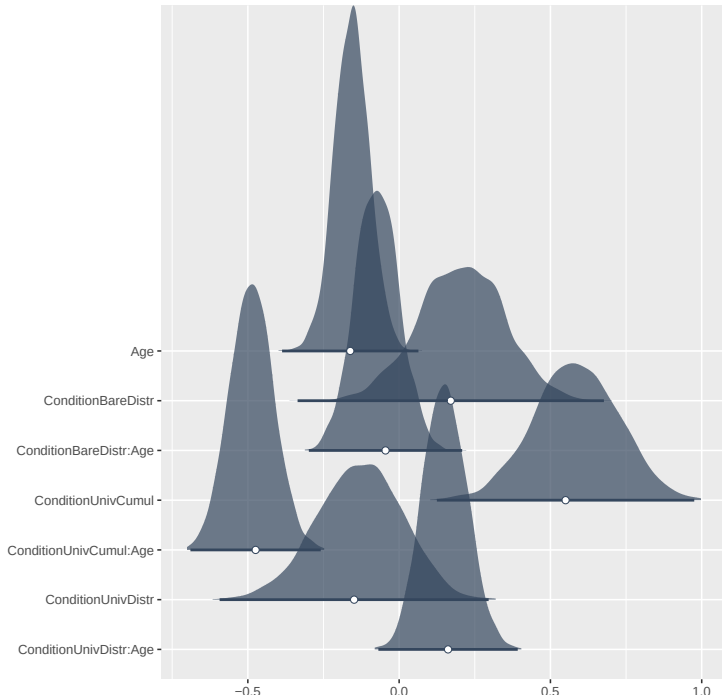
Inferential statistics

B) children

- the cumulative interpretation of bareNPs is strongly acceptable
 - ref. level – the intercept: $\hat{\mu} = 1.08$, $CI = [0.86, 1.30]$
- and doesn't change during acquisition
 - the main effect of AGE is non-credible: $\hat{\mu} = -0.16$, $CI = [-0.27, -0.04]$, 14.42% in ROPE
- posterior samples graph for adults Bayes model on the next slide

report

Posterior Samples



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describe_posterior(model_bayes2)
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Summary of Posterior Distribution

Parameter	Median	95% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	1.08	[0.86, 1.30]	100%	[-0.10, 0.10]	0%	1.001	1953.00
ConditionBareDistr	0.21	[-0.07, 0.47]	92.55%	[-0.10, 0.10]	21.21%	1.000	2164.00
ConditionUnivCumul	0.58	[0.31, 0.85]	100%	[-0.10, 0.10]	0%	1.001	2252.00
ConditionUnivDistr	-0.13	[-0.42, 0.13]	82.15%	[-0.10, 0.10]	40.29%	1.001	2282.00
Age	-0.16	[-0.27, -0.04]	99.72%	[-0.10, 0.10]	14.42%	1.001	1947.00
ConditionBareDistr:Age	-0.07	[-0.21, 0.07]	84.82%	[-0.10, 0.10]	64.03%	1.001	2178.00
ConditionUnivCumul:Age	-0.49	[-0.63, -0.34]	100%	[-0.10, 0.10]	0%	1.001	2304.00
ConditionUnivDistr:Age	0.15	[0.01, 0.30]	98.65%	[-0.10, 0.10]	22.82%	1.001	2290.00

1. Problém + plán: navázáním na předchozí lingv. literaturu: jak probíhá akvizice univerzálních kvantifikátorů a distributivity v češtině a jak se liší od akvizice holých NP v plurálu
 - nový jazykový vzorek (čeština) a nový typ výrazů (holé NP v plurálu)
2. Data + analýza: potvrzení dřívějších zjištění o relativně pozdním osvojení správné distributivní interpretace univerzálních kvantifikátorů (9–11 let)
 - Závěr: nové zjištění ohledně přetrvávajícího distributivního čtení holých NP v plurálu (i v 11 letech)
 - nový poznatek z češtiny → odpovídající obecným principům akvizice sémantiky a pragmatiky
 - využitelné jak v dalším výzkumu akvizice, tak i jako případná témata dizertačních a magisterských prací

3. predikce pro další výzkum:

- specificky české jazykové prostředky – např. genericita a její akvizice (roztřídění sémantických a pragmatických faktorů, které se na ní podílejí)
- klasický rekurzivní Problem, Plan, Data, Analysis, and Conclusion (PPDAC, viz Spiegelhalter (2019)) cyklus.

Illustrative example from Koster et al. (2018)

- (12) Elke jongen wast een boot.
Each boy.SG wash.SG a boat.SG
'Each boy is washing a boat.'



Fig. 6. Collective

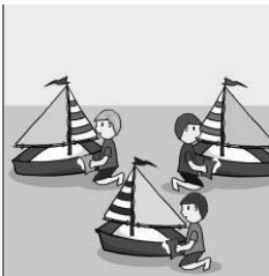


Fig. 7. Distributive

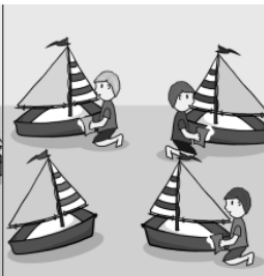


Fig. 8. Extra Object

Obrázek 9: Example image conditions from Koster et al. (2018)

Errors made by children:

- in Dutch (and all other observed languages)
 - until age 8 (and sometimes after)
1. children (correctly) accept *elke* with distributive interpretation
 2. children (incorrectly) accept *elke* with collective interpretation
 3. children (incorrectly) reject *elke* with scenes including extra object (boat)

- non-trivial error: it is general lack of understanding of distributivity, collectivity and cumulativity (not only with universal quantifiers)
- universal quantifier is one of the functional words: essential parts of the human grammar machinery

The acquisition errors I: spreading errors

- recall: children (≤ 9) incorrectly reject universal quantifier sentences with extra objects
- well-documented (Crain et al. 1996; Drozd 2001; Roeper et al. 2006)
- theoretical explanation (Geurts 2003): children interpret strong quantifiers as weak ones

Strong vs. weak quantifiers

- distinction already from Barwise and Cooper (1981)
- linguistic test: appearance in existential constructions:

- (13) a. There are {some/three boys } in the garden.
 b. There is {#each/every boy } in the garden.

- logical difference: weak quantifiers are symmetrical and strong quantifiers are relational

- (14) a. Some boys dance.
(i) $\exists x[\mathbf{boy}(x) \wedge \mathbf{dance}(x)] = \exists x[\mathbf{dance}(x) \wedge \mathbf{boy}(x)]$
b. Each boy dances.
(i) $\forall x[\mathbf{boy} \rightarrow \mathbf{dance}(x)] \neq \forall x[\mathbf{dance}(x) \rightarrow \mathbf{boy}(x)]$

- Geurts (2003): tripartite structure of strong quantifiers is more cognitively demanding
 - children interpret all quantifiers as weak until they acquire the adult stage

- Geurts (2003): children make spreading errors since they interpret the restrictor by pragmatics

- (15)
- | | | |
|----|---|-----------------|
| a. | Each [_{restrictor} boy] [_{nuclear scope} dances]. | |
| b. | Each [_{restrictor} boy] is washing a boat. | correct |
| c. | Each [_{restrictor} boat] is washed by a boy. | spreading error |

Musolino (2009): 4-6 years old English speaking children

- truth value judgment task (sentence + picture) - children incorrectly interpret *each*:

(16) Three boys are holding each balloon.

- against the picture on the next slide (90% acceptance vs. 31% acceptance in adults control group)



Obrázek 10: False distributive interpretation

- Musolino's explanation: wrong identification of the restrictor like in (17)

(17) Each of the three boys is holding a balloon.

Problems of Musolino's account

- the mechanism by which children identify the restrictor is not clear
- the expectation: the timing of distributivity and spreading errors should coincide
 - but this is not the case: Koster et al. (2018)
 - spreading errors (Dutch children) disappear after 9 years
 - but in this age (until 11) children still don't acquire the correct interpretation of $\forall$

Another explanation

- building on Dotlacil (2010): acquisition studies Pagliarini et al. (n.d.), Koster et al. (2018)
- conversational implicature explanation (after Horn (2001))
 - (18-a): non-ambiguously distributive
 - (18-b): principally ambiguous but (18-a) is logically stronger (asymmetrically entails) and non-ambiguous
 - cooperative speaker knowing that (18-a) would be true would use it
 - → (18-a) is false and (18-b) has only the non-distributive interpretation

- (18)
- a. Each boy drew two animals.
 - b. Two boys drew two animals.

- Barwise, Jon, and Robin Cooper. 1981. "Generalized Quantifiers and Natural Language." *Linguistics and Philosophy*, 159219.
- Crain, Stephen, Rosalind Thornton, Carole Boster, Laura Conway, Diane Lillo-Martin, and Elaine Woodams. 1996. "Quantification Without Qualification." *Language Acquisition* 5 (2): 83153.
- Dočekal, Mojmír, and Žaneta Šulíková. 2024. "The Distributive and Cumulative Readings Acquisition: Experimental Evidence from Czech." *Language Use and Linguistic Structure. Proceedings of the Olomouc Linguistic Colloquium 2023* 10: 193.

- Dotlacil, Jakub. 2010. *Anaphora and Distributivity: A Study of Same, Different, Reciprocals and Others*. Netherlands Graduate School of Linguistics.
- Drozd, Kenneth F. 2001. "12 Children's Weak Interpretations of Universally Quantified Questions." *Language Acquisition and Conceptual Development* 3: 340.
- Geurts, Bart. 2003. "Quantifying Kids." *Language Acquisition* 11 (4): 197218.
- Goodrich, Ben, Jonah Gabry, Imad Ali, and Sam Brilleman. 2022. *Rstanarm: Bayesian Applied Regression Modeling via Stan*.
<https://mc-stan.org/rstanarm/>.

Grice, Herbert P. 1975. "Logic and Conversation." In *Speech Acts*. Brill.

Horn, L. R. 2001. *A Natural History of Negation*. David Hume Series.
CSLI. <https://books.google.cz/books?id=hBFtAAAAIAAJ>.

Katsos, Napoleon, and Dorothy VM Bishop. 2011. "Pragmatic Tolerance:
Implications for the Acquisition of Informativeness and Implicature."
Cognition 120 (1): 67–81.

Koster, Anna de, Jennifer Spenader, and Petra Hendriks. 2018. *Are Children's
Overly Distributive Interpretations and Spreading Errors Related?*

- Musolino, Julien. 2009. "The Logical Syntax of Number Words: Theory, Acquisition and Processing." *Cognition* 111 (1): 2445.
- Noveck, Ira A. 2001. "When Children Are More Logical Than Adults: Experimental Investigations of Scalar Implicature." *Cognition* 78 (2): 165–88.
- Pagliarini, Elena, Gaetano Fiorin, and Jakub Dotlačil. n.d. *The Acquisition of Distributivity in Pluralities*.
- Papafragou, Anna, and Julien Musolino. 2003. "Scalar Implicatures: Experiments at the Semantics–Pragmatics Interface." *Cognition* 86 (3): 253–82.

Pinker, S. 1995. *The Language Instinct*. HarperPerennial.

<https://books.google.cz/books?id=00FrAAAAIAAJ>.

Roeper, Thomas, Uri Strauss, and Barbara Zurer Pearson. 2006. "The Acquisition Path of the Determiner Quantifier Every: Two Kinds of Spreading." *Papers in Language Acquisition, University of Massachusetts Occasional Papers UMOP* 34.

Sauerland, Uli, and Kazuko Yatsushiro. 2018. "The Acquisition of Disjunctions: Evidence from German Children." *Proceedings of Sinn Und Bedeutung* 21: 1065–72.

Šimík, Radek, and Christoph Demian. 2020. “Definiteness, Uniqueness, and Maximality in Languages with and Without Articles.” *Journal of Semantics* 37 (3): 311366.

Spiegelhalter, David. 2019. *The Art of Statistics: How to Learn from Data*. Basic Books.

Šulíková, Žaneta. 2023. “Distributivita, Kvantifikátory a Interpretace Jmenných Frází v Přirozeném Jazyce: Akviziční Experiment [Online].” Diplomová práce, Masarykova univerzita, Filozofická fakulta, Brno. <https://is.muni.cz/th/kcalx/>.

Syrett, Kristen. 2018. "The Historical Emergence and Current Study of Semantics in Acquisition." In *Semantics in Language Acquisition*. John Benjamins.